

EC-390: Digital Signal Processing

LECTURE NO 1: *Introduction to Digital Signal Processing*

Engr. Tehseen Ahsan
Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan

Organization of Lecture

1. Introduction.
2. Advantages.
3. Disadvantages.
4. Applications.
5. Signal.
6. Discrete Time System
7. Analysis of Discrete Time System.
8. Filters.

1. Introduction (1/2)

- **D**igital **S**ignal **P**rocessing (DSP) refers to processing of signals by digital systems like Personal Computers (PC) and systems designed using digital Integrated Circuits (ICs), microprocessors and microcontrollers.
- The basic components of a DSP system are shown in fig 1. The **DSP system** involves conversion of analog signal to digital signal, then processing of the digital signal by a digital system and then conversion of the processed digital signal back to analog signal.

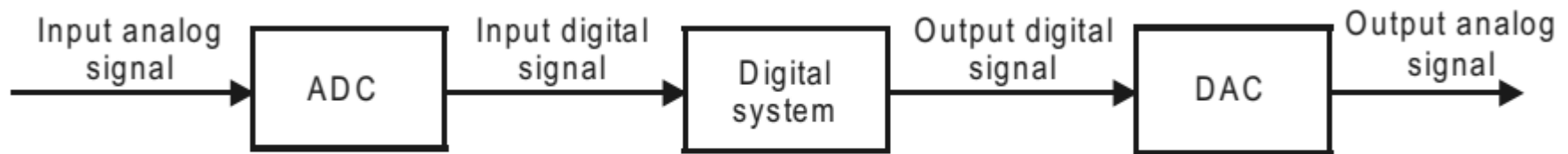


Fig 1 : Basic components of a DSP system.

1. Introduction (2/2)

- The real-world signals are analog, and only for processing by digital systems, the signals are converted to digital. For conversion of signals from analog to digital, an ADC (Analog to Digital Converter) is employed. The various steps in analog to digital conversion process are sampling and quantization of analog signals, and then converting the quantized samples to suitable binary codes. The digital signals in the form of binary codes are fed to digital system for processing, and after processing, it generates an output digital signal in the form of binary codes. The output analog signal is constructed from the output binary codes using a DAC (Digital to Analog Converter).
- The processing of signals are basically spectrum analysis to determine the various frequency components of a signal and filtering the signal to extract the required frequency component of the signal.
- The digital system can be a specially designed programmable hardware for DSP or an algorithm/software running on a general purpose digital system like Personal Computer (PC).

2. Advantages

- Some of the advantages of digital processing of signals are:
 - i. The digital hardware are compact, reliable, less expensive, and programmable.
 - ii. Since the DSP systems are programmable, the performance of the system can be easily upgraded/modified.
 - iii. By employing high speed, sophisticated digital hardware higher precision can be achieved in processing of signals.
 - iv. The digital signals can be permanently stored in magnetic media so that they are transportable and can be processed in non-real-time or off-line.

3. Disadvantages

- i. As we know that ADC and DAC converters are used in DSP, this makes the digital system more complex and also time for conversion is more.
- ii. Sampling leads to loss of information.
- iii. Limited speed of operation.

4. Applications

DSP is majorly used in the fields of:

- i. Radar.
- ii. Voice recognition.
- iii. Speech processing.
- iv. Image processing.
- v. Consumer electronics.
- vi. Biomedical signal processing.
- vii. Data compression and image compression.

6. Signal

- Any physical phenomenon that conveys or carries some information can be called a **signal**. The music, speech, motion pictures, still photos, heart beat, etc., are examples of signals that we normally encounter in day-to-day life.
- When a signal is defined continuously for any value of an independent variable, it is called an analog or **continuous signal**. It is denoted by “ $x(t)$ ”. Most of the signals encountered in science and engineering are analog in nature.
- When a signal is defined for discrete intervals of an independent variable, it is called a **discrete signal**. It is denoted by “ $x(n)$ ”.
- The quantized and coded version of the discrete time signals are called **digital signals**. In digital signals the value of the signal for every discrete time “ n ” is represented in binary codes. The process of conversion of a discrete time signal to digital signal involves quantization and coding.

7. Discrete Time System (1/2)

- A discrete time system is denoted by the letter $\mathcal{H}$. The input of discrete time system is denoted as “ $x(n)$ ” and the output of discrete time system is denoted as “ $y(n)$ ”. The diagrammatic representation of a discrete time system is shown in fig 1.

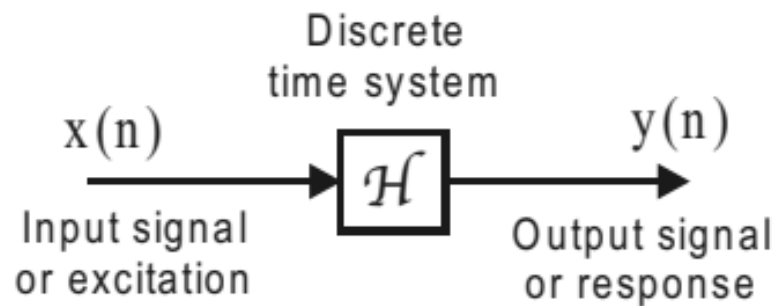


Fig 1 : Representation of discrete time system.

7. Discrete Time System (2/2)

- The operation performed by a discrete time system on input to produce output or response can be expressed as:

Response, $y(n) = \mathcal{H}\{x(n)\}$

where, $\mathcal{H}$ denotes the system operation (also called system operator).

8. Analysis of Discrete Time System (1/2)

- Mostly, the discrete time systems are designed for analysis of discrete time signals. Physically, the discrete time systems are realized in time domain. In time domain, the discrete time systems are governed by difference equations. The analysis of discrete time signals and systems in time domain involves solution of difference equations. The solution of difference equations are difficult due to assumption of a solution and then solving the constants using initial conditions.
- In order to simplify the task of analysis, the discrete time signals can be transformed to some other domain, where the analysis may be easier. One such transform exists for discrete time signals is Z-transform. The Z-transform, will transform a function of discrete time “ n ” into a function of complex variable “ z ”

8. Analysis of Discrete Time System (2/2)

- The ratio of Z-transform of output and Z-transform of input is called *transfer function* of the discrete time system. It is denoted by $H(z)$ and $H(z) = Y(z)/X(z)$
- The inverse Z-transform of transfer function is the *impulse response* of the system. It is denoted by $h(n)$ and $h(n) = Z^{-1}\{H(z)\}$ or $h(n) = Z^{-1}\{Y(z)/X(z)\}$.
- The *Fourier transform*, will transform a function of discrete time “n” into a function of frequency “W”. Therefore, Fourier transform of a discrete time signal will transform the discrete time signal into frequency domain signal. The Fourier transform of the discrete time signal, is also called *frequency spectrum* of the discrete time signal. The Fourier transform of the impulse response of a system is called *frequency response* of the system.

9. Filters

- The filters are frequency selective devices. The two major types of digital filters are FIR (**F**inite **I**mpulse **R**esponse) and IIR (**I**nfinite **I**mpulse **R**esponse) filters.

Note : Mathematically, the filter design is design of transfer function of the filter.

References: Textbooks

1. *“Discrete-Time Signal Processing”, International Edition, by Oppenheim, Alan V; Schafer, Ronald W, (2013), Pearson.*
2. *“Digital Signal Processing Using Matlab”, 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related documents from open source, mainly internet.*